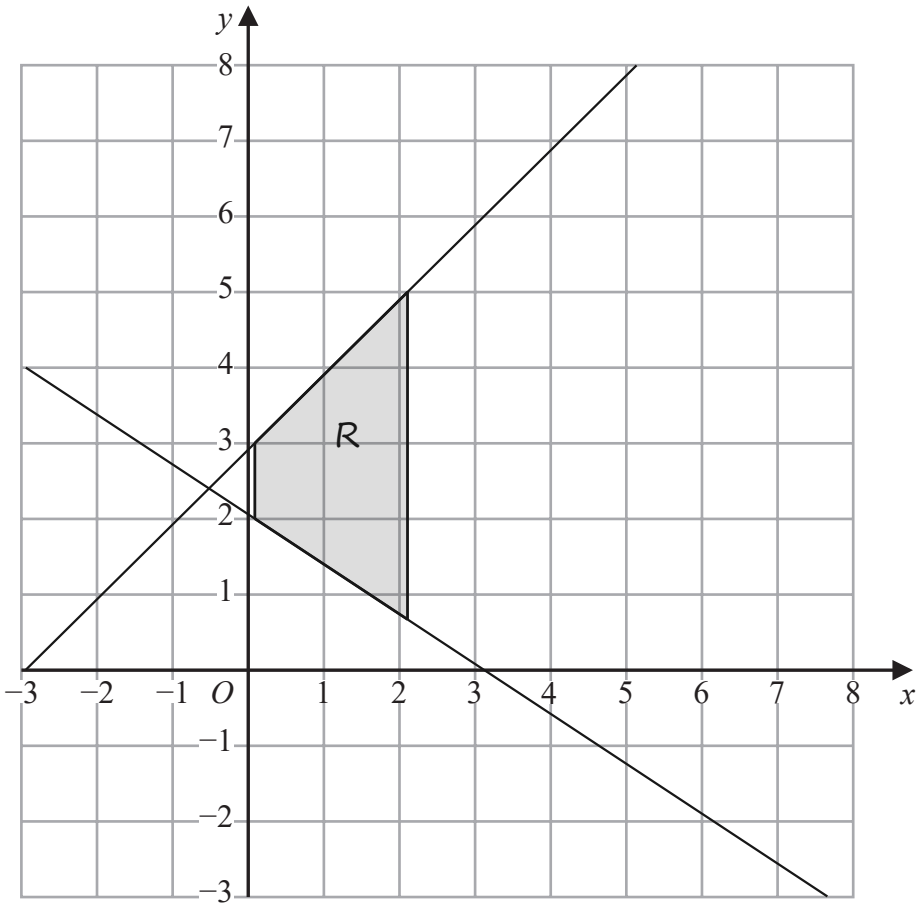
13 (a) On the grid show, by shading, the region that satisfies all these inequalities.

$x \geqslant 0$ $x \leqslant 2$ $y \leqslant x + 3$ $2x + 3y \geqslant 6$

Label the region **R**.



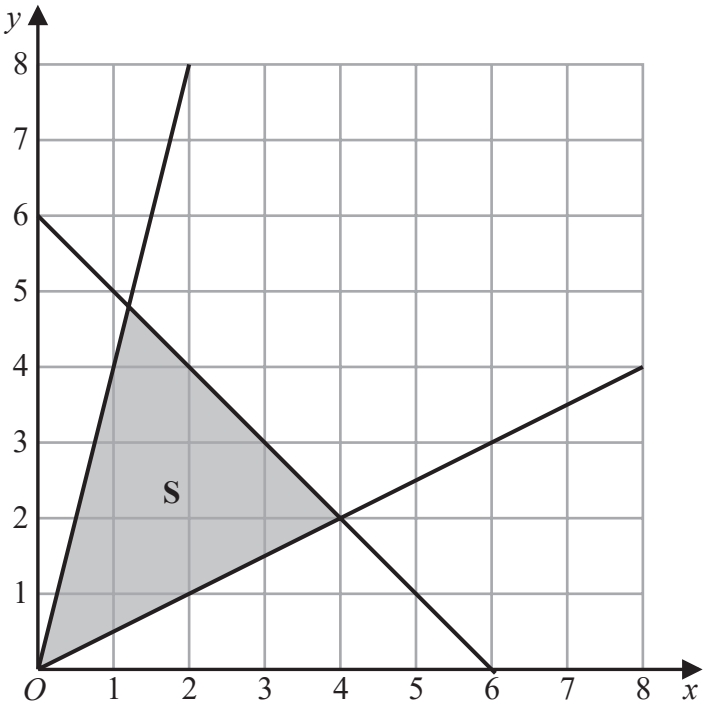
(4)

4 marks total
x=2, y=x+3
2x+3y=6
Correct side of lines
Region R

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(b) The diagram below shows the region **S** that satisfies the inequalities

$$y \leqslant 4x \qquad y \geqslant \frac{1}{2}x \qquad x + y \leqslant 6$$



Geoffrey says that the point with coordinates (2, 4) does not satisfy all the inequalities because it does not lie in the shaded region.

Is Geoffrey correct?
You must give a reason for your answer.

No: (2,4) has 4 ≤ 4×2 and 4 ≥ ½×2.

Also 2 + 4 = 6, so it lies on the included boundary.

1 mark
Boundary equality included
No, Geoffrey is wrong

(1)

(Total for Question 13 is 5 marks)

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